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10.2 Volume

Volume measures the space enclosed by a solid, in cubic units. Prisms and cylinders use $V = Bh$ (base area times height); pyramids and cones use exactly $\frac{1}{3}$ of that, $V = \frac{1}{3}Bh$, because a pyramid/cone fills only a third of the prism/cylinder that encloses it; a sphere uses $V = \frac{4}{3}\pi r^3$. As with surface area, the height needed in the formula is the solid's *true vertical height*, which sometimes must be recovered from a slant height using the Pythagorean Theorem. This worksheet also covers composite solids, cone-vs-cylinder comparisons, reverse problems, and real-world capacity/unit-conversion problems.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Find the volume of a rectangular prism $7 \times 4 \times 3$.

2. Find the volume of a cylinder with radius 6, height 5.

3. Square pyramid, base edge 9, height 4. Find volume.

4. Cone with radius 4, height 9. Find volume.

5. Find the volume of a sphere with radius 9.

6. Find the volume of a sphere with diameter 8.

7. Find the volume of a cube with edge 8.

8. Cylinder with radius 2, height 10. Find volume.

9. Cone with radius 5, height 12. Find volume.

10. Square pyramid, base edge 10, slant height 13. Find the height, then the volume.

11. A silo is a cylinder (radius 3, height 8) topped by a hemisphere of the same radius. Find the total volume.

12. A cone has radius 6 and volume 96π. Find its height.

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13. A cone and a cylinder share the same base and height. What fraction of the cylinder's volume does the cone hold?

14. A cylindrical tank has diameter 6 m and height 4 m. Find its volume in m^3 (exact form), then estimate its capacity in liters.

15. A sphere is inscribed in a cube of edge length 10 so that it touches the center of all six faces (the sphere's radius equals half the cube's edge). Find the exact volume *inside the cube but outside the sphere*.

✓ Answer Key

1. 84
2. 180π
3. 108
4. 48π
5. 972π
6. $\dfrac{256\pi}{3}$
7. 512
8. 40π
9. 100π
10. $h = 12;\ V = 400$
11. 90π
12. $h = 8$
13. $\dfrac{1}{3}$
14. $36\pi\ \text{m}^3 \approx 113{,}100\ \text{L}$
15. $1000 - \dfrac{500\pi}{3}$